MILITARY SPECIFICATION

LIGHTS: EXTENSION, FLUORESCENT, EXPLOSION PROOF

1. SCOPE

1.1 Scope. This specification covers one type of explosion proof fluorescent portable extension light.

1.2 Classification. Lights shall be of the following classes as specified (see 6.2):

Class I - High output light - 10 watts (W).

Class II - Low output light - 8W.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

Federal

| | |
|---|---|
| J-C-580 | Cord, Flexible, And Wire, Fixture, (Electrical, 0- To 600-Volt Service) |
| PPP-B-601 | Boxes, Wood, Cleated-Plywood |
| PPP-B-636 | Box, Fiberboard |
| PPP-B-640 | Boxes, Fiberboard, Corrugated, Triple-Wall |
| PPP-T-60 | Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging |

Military

| | |
|---|---|
| MIL-P-116 | Preservation, Methods Of |
| MIL-V-173 | Varnish, Moisture-And-Fungus-Resistant (For The Treatment Of Communications, Electronic, And Associated Electrical Equipment) |
| MIL-D-1000 | Drawings, Engineering And Associated Lists |
| MIL-L-10547 | Liners, Case, And Sheet, Overwrap; Water-Vaporproof Or Waterproof, Flexible |



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STANDARDS

Military

| | |
|---|---|
| MIL-STD-129 | Marking For Shipment And Storage |
| MIL-STD-130 | Identification Marking Of US Military Property |
| MIL-STD-143 | Specifications And Standards, Order Of Precedence For The Selection Of |
| MIL-STD-1186 | Cushioning, Anchoring, Bracing, Blocking, And Waterproofing; With Appropriate Test Methods |

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNDERWRITERS' LABORATORIES, INCORPORATED

List of Inspected Electrical Equipment

Standard 781 Portable Fixtures, Electrical Lighting For Use In Hazardous Locations

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 161 Sixth Avenue, New York NY 10003; 207 East Ohio Street, Chicago IL 60611; or 1655 Scott Lane, Santa Clara CA 95050.)

3. REQUIREMENTS

3.1 Certification of approval by Underwriters' Laboratories, Incorporated. Prior to submission of items for acceptance, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that Underwriters' Laboratories, Incorporated has approved the item as an electric lighting fixture suitable for use in Class I, Group D hazardous locations.

3.2 Components. The extension light shall consist of:

| Item No. | Quantity | Description | See Requirement |
|---|---|---|---|
| 1 | 1 | Fixture Assembly | 3.6.1 |
| 2 | 1 | Extension Cord | 3.6.2 |



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3.3 Selection of specifications and standards. Specifications and standards for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitations for bids, there are no suitable standard parts. In any case, commercial utility parts such as screws, bolts, nuts and cotter pins having suitable properties may be used provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and on the contractor's drawings.

3.3.2 Standard parts. With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Materials.

3.4.1 Fungus-proof materials. Materials that are nutrients for fungi shall not be used where it is practical to avoid them. Where used and not hermetically sealed, they shall be treated with a fungus-resistant varnish in accordance with MIL-V-173. However, if they will be used in a hermetically sealed enclosure, fungus-resistant varnish will not be necessary.

3.4.2 Metallic parts. Metallic parts for the light shall be made of metals specified. Where ferrous metals are used, they shall be suitably protected against corrosion.

3.4.3 Metals. All exposed metal parts shall be of nonsparking metal.

3.5 Design and construction. The extension light shall be so constructed that no parts will work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, installation, and service.

3.5.1 Adjustments and repairs. The extension light shall be so constructed that lamp replacements may be easily made without the use of special tools, and all other adjustments and repairs may be easily made by the personnel of operating units and overhaul bases.

3.6 Details of components.

3.6.1 Fixture assembly. Each fixture assembly shall consist of a guard, clear glass globe, plastic or metal handle and an interior parts subassembly assembled together to form an explosion proof light. Class 1 lights shall be designed to accommodate a 10W, 120 volts (V) U-shape fluorescent lamp with a 4-pin base and Class II lights shall be designed to accommodate a standard 8W,



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120V T-8 type fluorescent lamp. The lamp assemblies shall conform to all requirements of the Underwriters' Laboratories Standard 781 for use in Class I, Group D, hazardous locations. The design of the fixture assembly shall be such as to permit relamping and replacement of starter without the aid of special tools.

3.6.1.1 Handle. The handle shall be cast or fabricated of nonferrous materials. It shall be designed with the cord entrance at the outer end, and with a cord connector suitable to accept the bayonet type contacts of a male connector attached to the interior parts subassembly.

3.6.1.2 Guard. The guard shall be cast or fabricated of nonferrous materials. One end shall be internally threaded to receive the handle and the other end shall accommodate the hook.

3.6.1.3 Interior parts subassembly. The interior parts subassembly shall consist of a fabricated metal frame on which shall be mounted the lamp socket, lamp starter socket, lamp ballast and male connector for making electrical connection to the handle. The design shall be such that improper assembly to the handle is not possible, and that correct orientation and polarity of electrical connection is assured.

3.6.1.4 Hook. The hook shall be located on the outer end of the guard. The design shall be such that no part of the hook will extend outside of the projected cylindrical surface of the guard.

3.6.1.5 Socket. The socket(s) shall be of a design which will accommodate the lamp specified in 3.6.1. They shall be of porcelain or other material suitable for high-temperature operation.

3.6.2 Extension cord. The extension cord shall be equipped with 75 ±1 feet of No. 16, 3-conductor, type SO flexible cord in accordance with J-C-580. One conductor of the cord shall be grounded to the fixture assembly and connected to the ground terminal of the plug. The extension cord shall be supplied less the connector plug.

3.7 Interchangeability. All parts functionally and mechanically interchangeable shall be identified by one part number. The item identification and part number requirements of MIL-D-1000 shall govern manufacturer's part number and changes thereto.

3.8 Dimensions. The overall length of the extension light, not including the extension cord, shall not be greater than 26 inches. The greatest diameter of the light shall be no greater than 3 1/2 inches.

3.9 Weight. The weight of the complete extension light, less extension cord and plug, shall be held to a minimum and shall not exceed 3 1/2 pounds.

3.10 Operation markings.



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